Introduction to Probability and Statistics
(EE 354 / CE 361)

Discrete RVs: Functions of RVs, Expected Value for Common RVs

Today's Agenda

- Expected Value
  - Motivation, definition and intuition
  - Bernoulli RV
  - Discrete Uniform RV

- Functions of RVs
  - Definition
  - PMF
  - Mean Value

Random Variable / Probability Mass Function

- Random Variable:

Random phenomenon represented by numerical values

- Probability Mass Function

Probability a random variable equals an outcome

Expectation/Expected Value/Mean of a Random Variable

Expectation

We define the **expected value** (also called the **expectation** or the **mean**) of a random variable X, with PMF p_X, by

$$\mathbf{E}[X] = \sum_x x p_X(x).$$

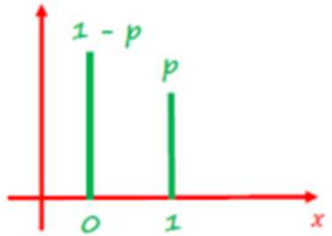
- Interpretations
  1. Weighted average of the possible values of X
  2. Center of gravity of the PMF of X
  3. Average of X in large number of repetitions of the experiment

Mean of a Bernoulli Random Variable

- Experiment
  - The toss of a coin which comes up heads with probability p and tails with probability $(1 - p)$.
- Sample Space
  - {Head, Tail}
- Random Variable
  - The RV takes value 1 if the outcome is head and takes the value 0 if the outcome is a tail.
- PMF
- Mean

$$X = \begin{cases} 1. & \text{if a head,} \\ 0. & \text{if a tail.} \end{cases}$$

$$p_X(k) = \begin{cases} p, & \text{if } k = 1, \\ 1 - p, & \text{if } k = 0. \end{cases}$$



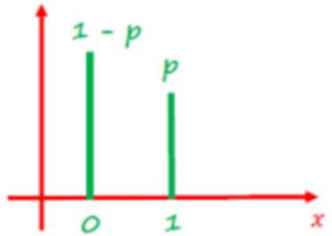
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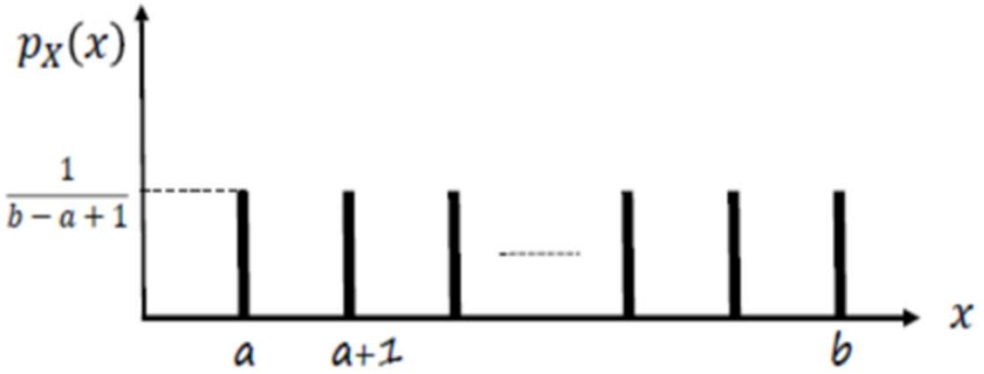
- Mean

$$\mathbf{E}[X] = \sum_x x\, p_X(x)$$

$$\mathbf{E}[X] = 1(p) + 0(1 - p) = p$$

Mean of a Discrete Uniform Random Variable

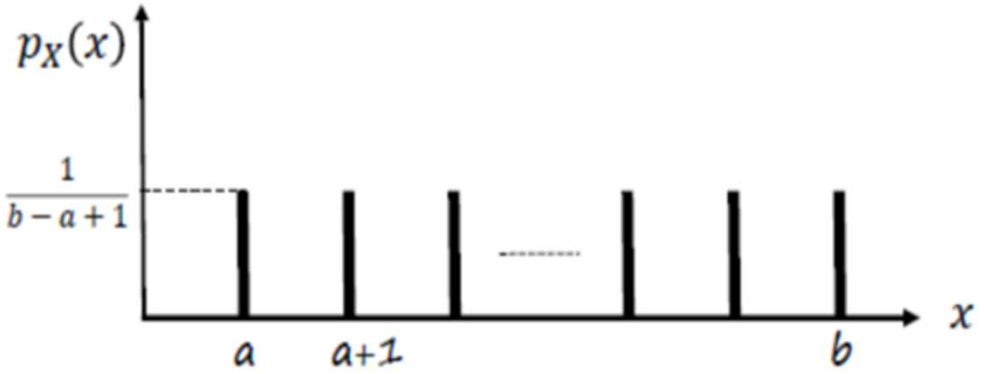
- Experiment
  - Pick one out of a range of contiguous integer values $(a, a+1, \ldots, b)$ with equal probability.
- Sample Space
  - $\{a, a+1, \ldots, b\}$
- Random Variable
  - For each integer in the sample space
    $X(w) = w$



- PMF $p_X(k) = \begin{cases} \frac{1}{b-a+1}, & \text{if } k = a, a+1, \ldots, b, \\ 0, & \text{otherwise.} \end{cases}$
- Mean

Mean of a Discrete Uniform Random Variable

- Experiment
  - Pick one out of a range of contiguous integer values $(a, a+1, \ldots, b)$ with equal probability.
- Sample Space
  - $\{a, a+1, \ldots, b\}$
- Random Variable
  - For each integer in the sample space
    $X(w) = w$



- PMF

$$p_X(k) = \begin{cases} \dfrac{1}{b-a+1}, & \text{if } k = a, a+1, \ldots, b, \\ 0, & \text{otherwise.} \end{cases}$$

- Mean

$$\mathbf{E}[X] = \frac{a+b}{2}$$

(since the PMF is symmetric around $(a+b)/2$)

Properties of Expectation/Mean

1. If

$$\forall w: X(w) \geq 0$$

Then

$$\mathbf{E}[X] \geq 0$$

2. If

$$\forall w: a \leq X(w) \leq b$$

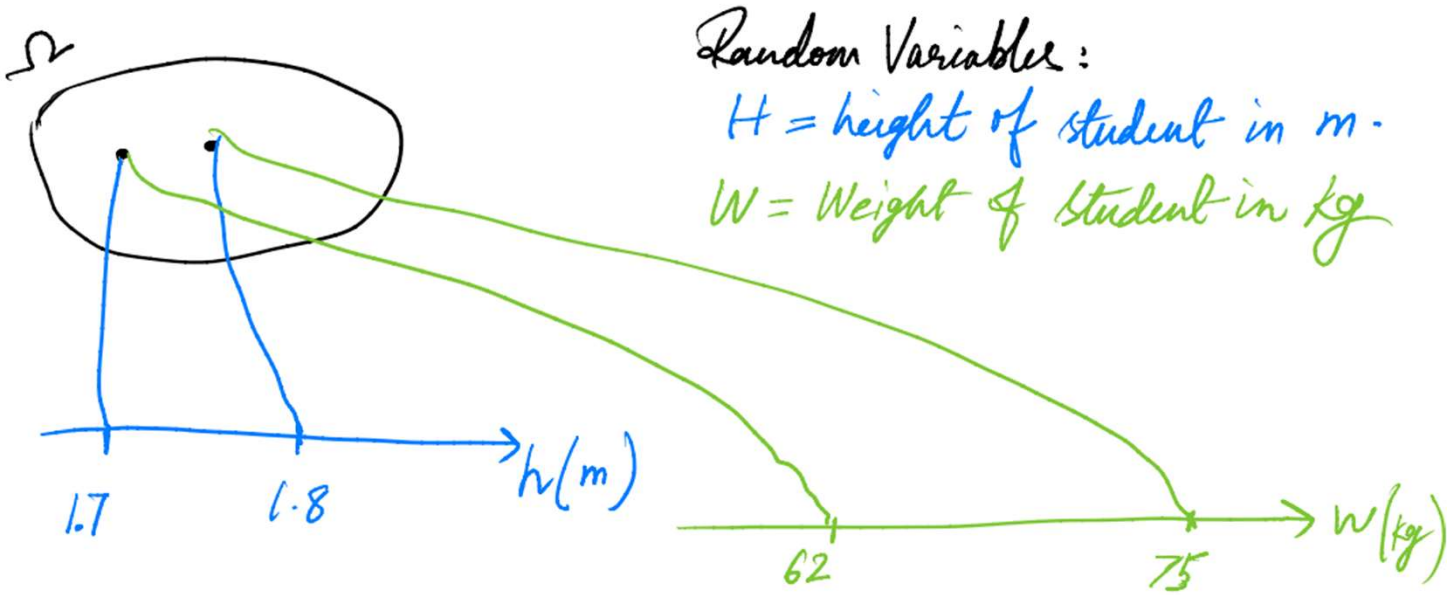
Then

$$a \leq \mathbf{E}[X] \leq b$$

Recall: Multiple Random Variables on the same Sample Space

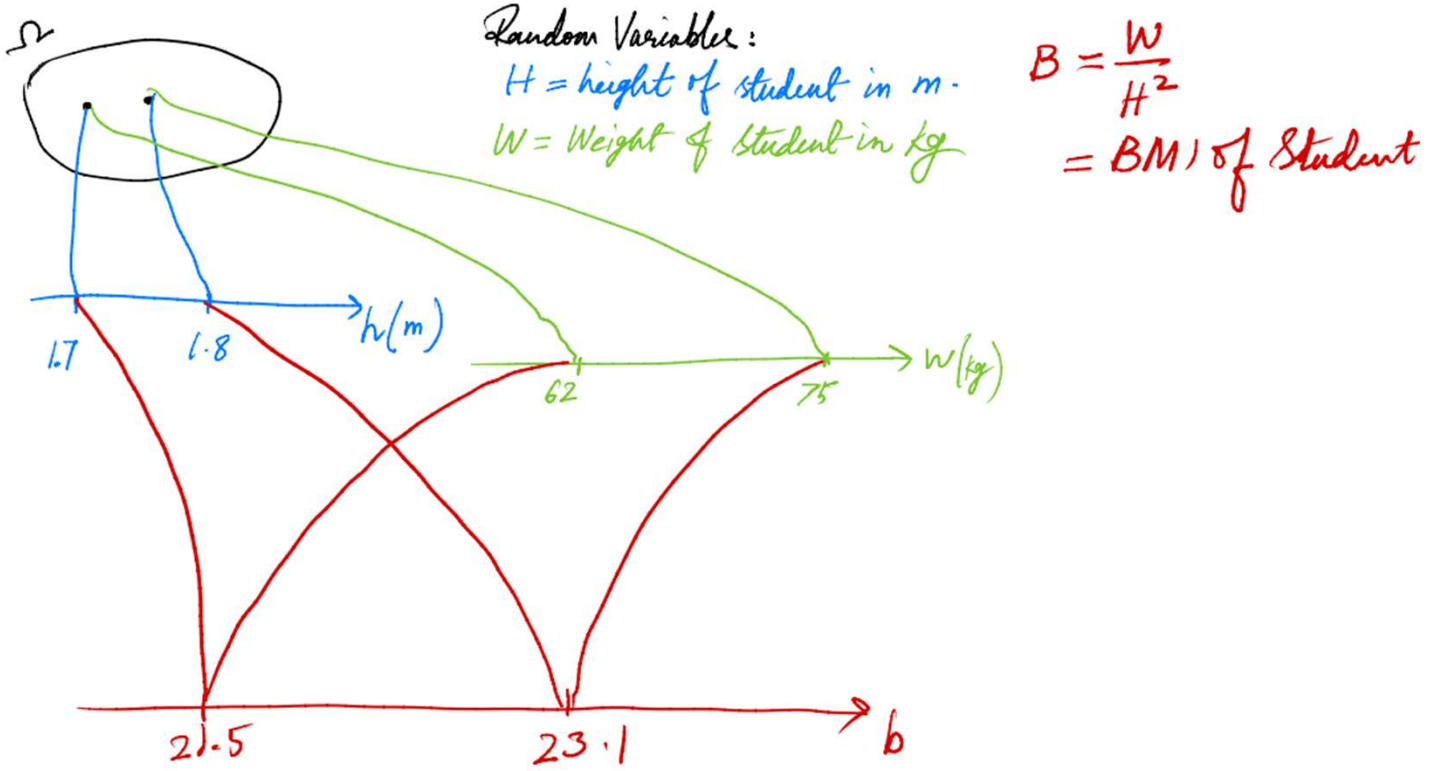
- We can have multiple random variables defined on the same sample space

- Example
  - Weight of a randomly selected student in the class
  - Height of a randomly selected student in the class



Recall: Functions of Random Variables

- The function of one or several random variables is also a random variable.



Functions of Random Variables

- The function of one or several random variables is also a random variable.

- If we define an RV as $X + Y$:

The RV takes on value $x + y$

when X takes on value x and Y takes on value y

PMF of a Function of a Random Variable

- As discussed earlier, if $Y = g(X)$ is a function of random variable X, then Y is also a random variable (with an associated PMF).

- If X is discrete with PMF p_X, then Y is also discrete and its PMF p_Y can be calculated using the PMF of X.

In particular, to obtain $p_Y(y)$ for any y, we add the probabilities of all values of x such that $g(x) = y$:

$$p_Y(y) = \sum_{\{x \mid g(x)=y\}} p_X(x).$$

Recall Example: Functions of Random Variables

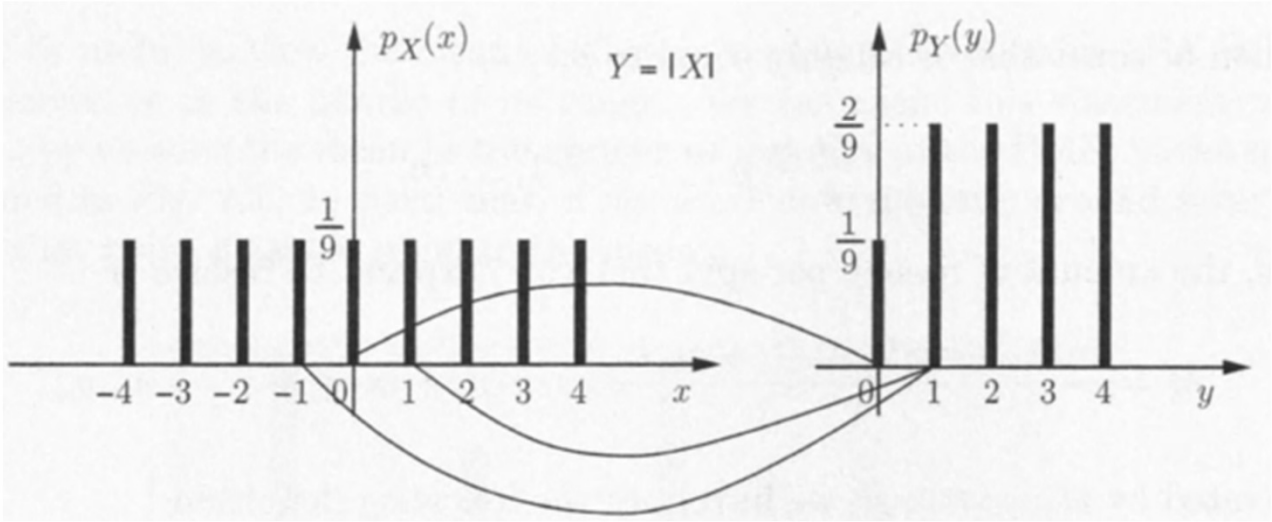
Example 2.1. Let $Y = |X|$ and let us apply the preceding formula for the PMF p_Y to the case where

$$p_X(x) = \begin{cases} 1/9. & \text{if } x \text{ is an integer in the range } [-4, 4], \\ 0. & \text{otherwise:} \end{cases}$$

What is $p_Y(y)$?

PMF of Y is

$$p_Y(y) = \begin{cases} 2/9. & \text{if } y = 1.2, 3, 4, \\ 1/9. & \text{if } y = 0. \\ 0. & \text{otherwise.} \end{cases}$$



Functions of Random Variables: Context/Utilization

X: Number of Red lights out of 5 signals that I encounter on my commute
T: Commute time

T = 20 + 2X

$$E[X] = \sum_x x \cdot p_X(x) = \mu_X \qquad E[T] = \sum_t t \cdot p_T(t) = \mu_T$$

- **Probability Mass Function (PMF) of X:**

$$p_X(x) = \begin{cases} \frac{1}{4}, & x = 0, 1 \\ \frac{1}{8}, & x = 2, 3, 4, 5 \end{cases}$$

- **Transformation Formula:**

$$T = g(X) = 20 + 2X$$

Functions of Random Variables: Context/Utilization

$$E[T] = \sum_{x=0}^{5} g(x)\, p_X(x) = \sum_{x=0}^{5} (20 + 2x)\, p_X(x)$$

- **For $x = 0$:** $(20 + 2(0)) \cdot \frac{1}{4} = 20 \cdot \frac{1}{4} = 5$
- **For $x = 1$:** $(20 + 2(1)) \cdot \frac{1}{4} = 22 \cdot \frac{1}{4} = 5.5$
- **For $x = 2$:** $(20 + 2(2)) \cdot \frac{1}{8} = 24 \cdot \frac{1}{8} = 3$
- **For $x = 3$:** $(20 + 2(3)) \cdot \frac{1}{8} = 26 \cdot \frac{1}{8} = 3.25$
- **For $x = 4$:** $(20 + 2(4)) \cdot \frac{1}{8} = 28 \cdot \frac{1}{8} = 3.5$
- **For $x = 5$:** $(20 + 2(5)) \cdot \frac{1}{8} = 30 \cdot \frac{1}{8} = 3.75$

$$E[T] = 5 + 5.5 + 3 + 3.25 + 3.5 + 3.75$$

$$\mathbf{E[T] = 24}$$

Expectation/Mean of a Function of a Random Variable

- We can extend the concepts of PMF and expectation developed so far for a single RV to multiple RVs

In particular, to obtain $p_Y(y)$ for any y, we add the probabilities of all values of x such that $g(x) = y$:

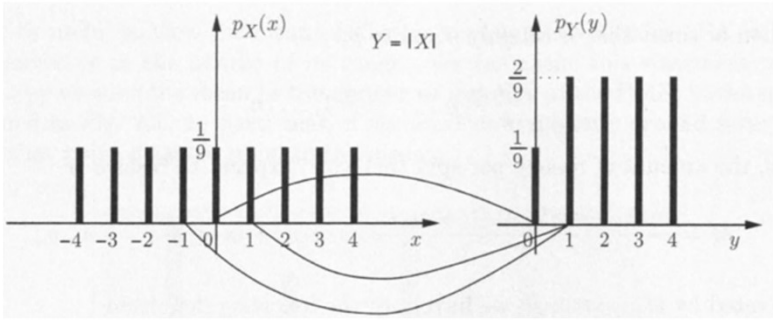
$$p_Y(y) = \sum_{\{x \mid g(x)=y\}} p_X(x).$$

- Now, the expectation of Y can be obtained by applying the definition to p_Y

$$\mathbf{E}[Y] = \sum_y y\, p_Y(y)$$

$$p_X(x) = \begin{cases} 1/9. & \text{if } x \text{ is an integer in the range } [-4, 4], \\ 0. & \text{otherwise:} \end{cases}$$

$$p_Y(y) = \begin{cases} 2/9. & \text{if } y = 1, 2, 3, 4, \\ 1/9. & \text{if } y = 0. \\ 0. & \text{otherwise.} \end{cases}$$



Expectation/Mean of a Function of a Random Variable

If $Y = g(X)$ is a function of random variable X,

- **The Harder Approach for Calculating Mean (Last slide):**
  - Calculate the PMF of Y using the PMF of X
  - Now, the expectation of Y can be obtained by applying the definition to p_Y: $\mathbf{E}[Y] = \sum_y y\, p_Y(y)$
- **An Easier Approach for Calculating Mean:**

Expected Value Rule for Functions of Random Variables

Let X be a random variable with PMF p_X, and let $g(X)$ be a function of X. Then, the expected value of the random variable $g(X)$ is given by

$$\mathbf{E}[g(X)] = \sum_x g(x) p_X(x).$$

  - Allows us to calculate the expectation of Y directly from the PMF of X

Expectation/Mean of a Function of a Random Variable

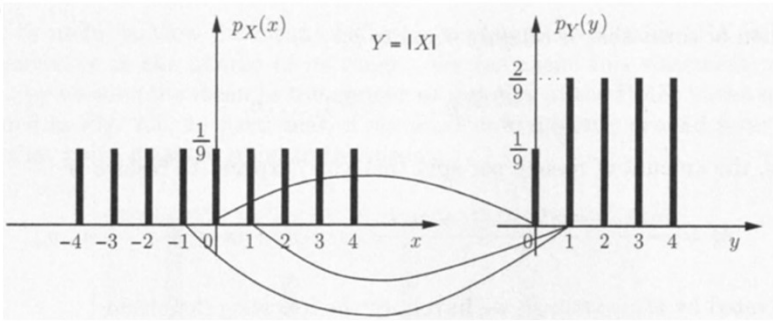
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Expected Value Rule for Functions of Random Variables: The Proof

Expected Value Rule for Functions of Random Variables

Let X be a random variable with PMF p_X, and let $g(X)$ be a function of X. Then, the expected value of the random variable $g(X)$ is given by

$$\mathbf{E}\big[g(X)\big] = \sum_x g(x)p_X(x).$$

To verify this rule, we let $Y = g(X)$ and use the formula

$$p_Y(y) = \sum_{\{x \mid g(x)=y\}} p_X(x)$$

$$\begin{aligned}
\mathbf{E}\big[g(X)\big] &= \mathbf{E}[Y] \\
&= \sum_y y p_Y(y) \\
&= \sum_y y \sum_{\{x \mid g(x)=y\}} p_X(x) \\
&= \sum_y \sum_{\{x \mid g(x)=y\}} y p_X(x) \\
&= \sum_y \sum_{\{x \mid g(x)=y\}} g(x)p_X(x) \\
&= \sum_x g(x)p_X(x).
\end{aligned}$$

Mean of a Linear Function of a Random Variable

Let X be a random variable and let

$$Y = aX + b.$$

where a and b are given scalars. Then,

$$\mathbf{E}[Y] = a\mathbf{E}[X] + b,$$

$$\mathbf{E}[Y] = \sum_x (ax + b)\, p_X(x)$$

$$= a \sum_x x\, p_X(x) + b \sum_x p_X(x)$$

$$= a\, \mathbf{E}[Y] + b$$

Caution: unless g(X) is a linear function, it is not generally true that $\mathbf{E}[g(X)]$ is equal to $g(\mathbf{E}[X])$. In general, $\mathbf{E}[g(X)] \neq g(\mathbf{E}[X])$